

IndicRAG-QA: Evidence-Grounded Cross-Lingual Question Answering for Indic and Code-Mixed Languages

1. Motivation

Useful information in India is increasingly available through university regulations, government documents, public policies, educational material, FAQs, reports, and other digital resources. However, users may ask questions in English, an Indic language, or an Indic-English code-mixed form, while the underlying document may be written in another language. Traditional keyword search can fail when the query and document use different words or languages. Multilingual semantic representations and Retrieval-Augmented Generation (RAG) can bridge this gap by retrieving relevant evidence before generating an answer. The key challenge is to ensure that the final answer is supported by the retrieved document rather than produced from unsupported model knowledge.

2. Problem Statement

The project will develop an evidence-grounded cross-lingual question answering system. A user may submit a query in English, one Indian language, or a code-mixed form. The system must retrieve the most relevant passages from a document collection and generate an answer using only that evidence. It must also detect when sufficient evidence is unavailable and return an UNANSWERABLE response instead of hallucinating.

$$D = \{d_1, d_2, \dots, d_n\}, \quad R(q) = \{p_1, p_2, \dots, p_k\}, \quad A = f(q, R(q))$$

Central Research Question: How effectively can multilingual semantic retrieval and RAG improve question answering for Indic and code-mixed queries compared with lexical retrieval, while keeping generated answers grounded in evidence?

3. Proposed Scope

Students will use one public-information domain such as university regulations, government schemes, public policies, education, tourism, agriculture, public FAQs, or another instructor-approved domain. A collection supporting approximately 300–500 question-answer instances is recommended.

- Use English + at least one Indian language; include a meaningful subset of Indic-English code-mixed queries.
- Preprocess and segment documents into retrievable passages with document/passage identifiers.
- Implement a lexical retrieval baseline such as TF-IDF or BM25.
- Implement multilingual dense retrieval using IndicBERT, MuRIL, mBERT, XLM-R, multilingual Sentence-BERT, or another justified model.
- Develop a RAG-based answer generator that cites the supporting passage.
- Include both ANSWERABLE and deliberately UNANSWERABLE questions.

4. Suggested Technical Approach

The project should compare progressively stronger retrieval and question-answering pipelines rather than only connecting an LLM to a document store.

- Document Processing: cleaning, normalization, passage segmentation, tokenization, and metadata indexing.
- Lexical Retrieval: TF-IDF/BM25 with cosine or ranking-based relevance scoring.
- Semantic Retrieval: encode queries and passages into multilingual vector representations and retrieve Top-K passages.
- Hybrid Retrieval: combine lexical and semantic evidence, e.g., $\text{Score}(p) = \alpha \cdot \text{Score}_{\text{lexical}} + (1-\alpha) \cdot \text{Score}_{\text{semantic}}$.
- RAG Question Answering: provide Top-K passages to a transformer/LLM and instruct it to answer only from retrieved context.
- Answerability Detection: reject unsupported questions when evidence confidence is below a suitable threshold.

5. Required Modules:

- Module 1: TF-IDF/BM25 vs multilingual dense retrieval.
- Module 2: Compare monolingual, cross-lingual, and code-mixed queries.
- Module 3: Compare lexical, dense, and hybrid retrieval.
- Module 4: Direct LLM question answering vs retrieval-augmented question answering.
- Module 5: Hallucination/answerability test using intentionally unanswerable questions.
- Module 6: Qualitative analysis of at least 15 difficult cases involving code-mixing, spelling variation, named entities, numbers, dates, ambiguity, or semantically similar passages.

6. Evaluation

Retrieval and answer-generation components must be evaluated separately so that students can identify whether errors originate from retrieval or generation.

Component	Suggested Metrics
Retrieval	Recall@K, Precision@K, Hit Rate@K, Mean Reciprocal Rank (MRR)
Question Answering	Exact Match, token-level F1, semantic similarity, answer correctness
Answerability	Accuracy, Precision, Recall, F1-score, Confusion Matrix
Language Analysis	Separate results for English, Indic, cross-lingual and code-mixed queries

$$MRR = (1/N) \sum_i 1/\text{rank}_i$$

7. Expected Output

For each query, the system should ideally return:

- Detected query language/type: English / Indic / Code-Mixed.
- Answerability: ANSWERABLE / UNANSWERABLE.
- Generated answer and confidence score.
- Top supporting passage(s) and source document identifier.
- A short evidence-grounded explanation where feasible.

Example: Query: “Scholarship ke liye minimum eligibility kya hai?” → Answerability: ANSWERABLE → Answer + confidence + supporting passage + source. If evidence is absent, the system should return “Insufficient information available in the provided documents.”

8. Suggested Pipeline

Document Collection → Cleaning & Passage Segmentation → Lexical/Dense Indexing → Query Processing → Cross-Lingual Retrieval → Top-K Passages → RAG Answer Generation → Answerability Check → Evidence-Grounded Answer + Source

9. Recommended Research Extensions (Optional/ Bonus Work)

- Hindi-English code-mixed QA and query normalization.
- Query translation vs direct multilingual embeddings.
- Sparse-dense hybrid retrieval and transformer reranking.
- Confidence-based answer rejection and hallucination reduction.
- Citation verification and multi-document question answering.
- Comparison of small language models and larger instruction-tuned models.

10. Expected Research Contribution

The work should experimentally determine whether multilingual dense retrieval outperforms lexical retrieval, whether hybrid retrieval improves robustness, whether RAG reduces unsupported answers compared with direct LLM use, and whether code-mixed queries are more difficult than monolingual queries. The project must include analysis rather than only an application demo.

11. Deliverables Checklist

- Source Code (GitHub) with README and execution instructions.
- Research Paper (IEEE Style Word & PDF).
- Presentation Slides.
- Dataset/document collection description and question-answer set.
- Retrieval, QA, answerability, and error-analysis results.
- Working demonstration of the system.

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